Semester 5													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
OE-2	Open Elective 2 (NPTEL / SWAYAM / MOOC)	3	0	0	3	3	25	25	50	--	--	100
24HS301T	Engineering Economics	3	0	0	3	3	25	25	50	--	--	100
24CH301T	Mass Transfer - 1	3	0	0	3	3	25	25	50	--	--	100
24CH301P	Mass Transfer - 1 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH302T	Chemical Reaction Engineering - 1	3	0	0	3	3	25	25	50	--	--	100
24CH302P	Chemical Reaction Engineering - 1 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH303T	Process Equipment Design	3	0	0	3	3	25	25	50	--	--	100
24CH303P	Process Equipment Design Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-1	Programme Elective 1	Students must choose one course from the given basket of elective courses for study										
24CH331T	Petroleum Refining & Petrochemicals	3	0	0	3	3	25	25	50	--	--	100
24CH332T	Renewable Energy Engineering											
24CH333T	Sustainable and Green Chemistry											
	Total	**18**	**0**	**6**	**24**	**21**						

Semester 6													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
OE-3	Open Elective 3 (Inter-department, FoET)	3	0	0	3	3	25	25	50	--	--	100
24CH304T	Instrumentation and Process Control	3	0	0	3	3	25	25	50	--	--	100
24CH304P	Instrumentation and Process Control Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH305T	Chemical Reaction Engineering - 2	3	0	0	3	3	25	25	50	--	--	100
24CH305P	Chemical Reaction Engineering - 2 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH306T	Mass Transfer - 2	3	0	0	3	3	25	25	50	--	--	100
24CH306P	Mass Transfer - 2 Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-2	Programme Elective 2	Students must choose one course from the given basket of elective courses for study										
24CH334T	Piping Design	3	0	0	3	3	25	25	50	--	--	100
24CH335T	Corrosion Engineering											
24CH336T	Material Science & Engineering											
PE-3	Programme Elective 3	Students must choose one course from the given basket of elective courses for study										
24CH337T	Nano Technology & Energy Storage	3	0	0	3	3	25	25	50	--	--	100
24CH338T	Membrane Processes											
24CH339T	Environmental Engineering and Pollution Control											
	Total	**18**	**0**	**6**	**24**	**21**						

24CH304T					Instrumentation and Process Control					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the Laplace transformation method to solve differential equations.
- To provide a fundamental understanding of process control using the transfer function approach.
- To develop an understanding of various control actions such as P, PI and PID controllers.
- To understand the working of the most common instruments used in chemical industries.

Unit I: INTRODUCTION **Hours: 9**

Introduction and motivation for process control; block diagram representation; input-output model of a process; time and frequency domain representation of processes; Laplace transform, inverse Laplace transform and transfer functions; different types of input functions - delta, step, ramp, sinusoidal; defining piecewise continuous functions in terms of step inputs.

Unit II: PROCESS MODELING AND CONTROLLERS **Hours: 9**

First-order systems; linearization; second-order systems; stability analysis using system poles; Routh's stability test; feedback control; introduction to P, PI and PID controllers; servo and regulatory control; simplification of complex block diagrams; cascade control.

Unit III: FREQUENCY DOMAIN ANALYSIS **Hours: 9**

Frequency-domain analysis: amplitude ratio and phase; Bode stability criterion; Bode plots of simple gain, integrator, differentiator, first-order lag and first-order lead systems; controller tuning - stability-based tuning and direct synthesis methods of tuning; problem of inverse response and dead time in processes.

Unit IV: INSTRUMENTATION **Hours: 12**

Transducers and their dynamics; temperature measurement - resistance temperature device, thermocouples, thermistors; pressure measurement - manometers, diaphragms, capsules and bellows, Bourdon tubes; flow measurement; level measurement.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify an appropriate measuring instrument for quantities such as temperature, flow, pressure and liquid level.
CO2 - Understand process control systems from an input-output point of view using block diagrams.
CO3 - Apply Laplace transformations and inverse Laplace transformations to find the response of a process.
CO4 - Analyse the feedback control strategy using proportional (P), proportional-integral (PI) and proportional-integral-derivative (PID) control actions.
CO5 - Evaluate the stability of a process using system poles and Bode analysis.
CO6 - Build an appropriate controller by incorporating various controller tuning methods.

TEXT / REFERENCE BOOKS

1. Coughanowr, D. R., & LeBlanc, S. E. (2017). *Process systems analysis and control* (3rd ed.). McGraw Hill Education.
2. Stephanopoulos, G. (2008). *Chemical process control: An introduction to theory and practice*. Prentice-Hall India Learning.
3. Doebelin, E. O. (1990). *Measurement systems: Application and design* (4th ed.). McGraw-Hill.
4. Seborg, D. E., Edgar, T. F., Mellichamp, D. A., & Doyle, F. J. (2016). *Process dynamics and control* (4th ed.). John Wiley & Sons.

24CH304P					Instrumentation and Process Control Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To study the transient response of first and second-order systems for step and impulse changes in inputs.
- To study the interacting and non-interacting series connections of liquid-level systems.
- To study the working and calibration of RTDs, thermocouples and thermistors.
- To study pressure control and cascade control systems.

LIST OF EXPERIMENTS

1. Study of control valve characteristics: inherent and installed characteristics of control valves.
2. Study of hysteresis in control valves.
3. Study of the step response of a single-capacity system.
4. Study of the transient response of a non-interacting tank system.
5. Study of the transient response of an interacting tank system.
6. Temperature measurement using bimetallic and mercury-in-glass thermometers.
7. Temperature measurement using RTD, thermocouple and thermistors.
8. Study of a flapper-nozzle system.
9. Study of the step response of a second-order system using a mercury U-tube manometer.
10. Study of P, PI and PID controllers using a multi-process trainer.
11. Study of a cascade control system.
12. Study of a pressure control system.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify control valves using their flow characteristics.
CO2 - Understand the transient response of first and second-order processes.
CO3 - Determine the transfer function parameters of first and second-order processes.
CO4 - Calculate the gain of a flapper-nozzle system and of glass and bimetallic thermometers.
CO5 - Validate the experimental data from a thermocouple, RTD and thermistor against standard data.
CO6 - Adapt P, PI and PID control mechanisms to control level, flow and pressure.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Instrumentation and process control laboratory manual*. Pandit Deendayal Energy University.
2. Coughanowr, D. R., & LeBlanc, S. E. (2017). *Process systems analysis and control* (3rd ed.). McGraw Hill Education.
3. Stephanopoulos, G. (2008). *Chemical process control: An introduction to theory and practice*. Prentice-Hall India Learning.
4. Doebelin, E. O. (1990). *Measurement systems: Application and design* (4th ed.). McGraw-Hill.

24CH305T					Chemical Reaction Engineering - 2					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To accustom students to the concepts of non-ideality in reactor systems and the study of RTD.
- To provide details on modelling non-ideality using zero- and one-parameter models.
- To give a wholesome picture of catalysis, catalytic reactions and catalytic reactors.
- To give detailed insight into external and internal mass transfer effects in catalytic reactions.

Unit I: CATALYSIS AND CATALYTIC REACTORS **Hours: 12**

Basics of catalysis; catalytic reactions; steps in a catalytic reaction; synthesizing a rate law; mechanism and rate-limiting step; design of reactors for gas-solid reactions; heterogeneous data analysis for reactor design; catalyst deactivation; multiphase reactors; catalytic reactors for deactivating catalysts (temperature-time trajectories, circulating fluidized bed reactor, straight-through transport reactor).

Unit II: EXTERNAL DIFFUSION EFFECTS **Hours: 7**

External diffusion effects on heterogeneous reactions - mass transfer fundamentals; binary diffusion; external resistance to mass transfer; correlations for the mass transfer coefficient; the shrinking core model; simple scale concept of surface-to-volume ratio (cooling vs heat release) compared across different reactor sizes.

Unit III: INTERNAL DIFFUSION EFFECTS **Hours: 9**

Internal diffusion effects on heterogeneous reactions; diffusion and reaction in catalysts; Thiele modulus; Weisz-Prater criterion; Mears' criterion; internal effectiveness; overall effectiveness; falsified kinetics; estimation of mass transfer and reaction limited regimes; Hatta number; multiphase reactors with diffusion and reaction (fluidized bed reactor, slurry reactor and trickle bed reactor); reactivity hazards and thermal runaway.

Unit IV: NON-IDEALITY AND RTD MEASUREMENTS **Hours: 11**

Basics of non-ideality; distribution of residence times for chemical reactors - general characteristics; RTD measurement; reactor modelling with RTD; models for non-ideal reactors: zero-parameter models and one-parameter models; research studies on the modelling of real reactors.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Select a proper reaction mechanism and design the catalytic reactor by rate data analysis.
CO2 - Compare the reactor performance with and without internal or external mass transfer limitations.
CO3 - Appraise the performance of different reactors in multi-phase systems.
CO4 - Design, develop and modify reactor systems for specific real-life problems.
CO5 - Relate the basics of non-ideal reactors and the characteristics of residence time distributions.
CO6 - Interpret the RTD profiles and experimental data to model non-ideal reactors.

TEXT / REFERENCE BOOKS

1. Fogler, H. S. (2016). *Elements of chemical reaction engineering* (5th ed.). Prentice Hall.
2. Levenspiel, O. (1999). *Chemical reaction engineering* (3rd ed.). John Wiley & Sons.
3. Smith, J. M. (1988). *Chemical engineering kinetics* (3rd ed.). McGraw-Hill.
4. Schmidt, L. D. (1998). *The engineering of chemical reactions*. Oxford University Press.
5. Doraiswamy, L. K., & Sharma, M. M. (1984). *Heterogeneous reactions: Analysis, examples and reactor design* (Vols. 1-2). John Wiley & Sons.

24CH305P					Chemical Reaction Engineering - 2 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the non-ideality of different reactors by performing RTD studies.
- To study homogeneous reactions with and without catalysts to observe the change in reaction performance.
- To study the effect of mass transfer with and without reaction.
- To understand the modelling aspects using zero- and one-parameter models.

LIST OF EXPERIMENTS

1. Residence time distribution studies in a CSTR.
2. Residence time distribution studies in a PFR.
3. Residence time distribution studies in a PBR.
4. To study a non-catalytic homogeneous second order liquid phase reaction in a packed bed reactor.
5. To study the kinetics of dissolution of benzoic acid in water.
6. To study the catalytic decomposition of hydrogen peroxide in an adiabatic reactor.
7. Investigation of kinetics in a basket reactor.
8. Investigation of kinetics for dehydration of acetate in a recycle reactor.
9. Predict the conversion using the segregation model for different reactors.
10. Predict the conversion using the maximum mixedness model for different reactors.
11. Predict the conversion using the tanks-in-series model for different reactors.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the basics of non-ideality in reactors and their application in practice.
CO2 - Interpret the RTD data and evaluate different parameters for different reactors.
CO3 - Utilize the catalyst for the reaction and identify the catalytic effect on the performance of the reaction.
CO4 - Examine the effect of the presence or absence of mass transfer limitations on the reaction.
CO5 - Choose to model the RTD of a given reactor system and estimate the performance.
CO6 - Develop an ability to work in a group, learn report making and improve soft skills in presenting the findings.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Chemical reaction engineering laboratory manual*. Pandit Deendayal Energy University.
2. Fogler, H. S. (2016). *Elements of chemical reaction engineering* (5th ed.). Prentice Hall.
3. Levenspiel, O. (1999). *Chemical reaction engineering* (3rd ed.). John Wiley & Sons.
4. Smith, J. M. (1988). *Chemical engineering kinetics* (3rd ed.). McGraw-Hill.
5. Schmidt, L. D. (1998). *The engineering of chemical reactions*. Oxford University Press.

24CH306T					Mass Transfer - 2					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the principles of liquid-liquid and solid-liquid operations and the principle of vapour-liquid equilibria.
- To examine the importance of humidification and dehumidification for cooling purposes in industries.
- To identify engineering problems involving extraction, adsorption, humidification, dehumidification and drying.
- To apply the concept of number of transfer units in a cooling tower.

Unit I: DISTILLATION **Hours: 12**

Vapour-liquid equilibria; types of distillation; binary systems; McCabe-Thiele and Ponchon-Savarit methods; calculations with multiple feeds and withdrawals; azeotropic and extractive distillation; multicomponent distillation concept; distillation of heat-sensitive and flammable liquids: safety considerations in design and operation.

Unit II: EXTRACTION **Hours: 9**

Introduction to liquid-liquid extraction; liquid-liquid equilibrium; types of coordinate systems; solvent selection; stage-wise extraction; liquid-liquid extraction equipment; selection of extractors; principle and theory of solid-liquid extraction; types of operations; single and multistage operations; calculation of the required flow rate and mass transfer area for safe pre-discharge processing of released toxic materials.

Unit III: HUMIDIFICATION AND DEHUMIDIFICATION **Hours: 9**

Terminology and definitions; psychrometric charts and measurement of humidity; classification of cooling towers; cooling tower calculations; evaporative cooling; types of cooling towers and dehumidification methods.

Unit IV: DRYING AND CRYSTALLIZATION **Hours: 9**

Drying equilibrium and rate of drying; calculation of drying time; drying operations - batch and continuous; types of dryers and selection criteria. Crystallization: solid-liquid phase equilibrium; nucleation and crystal growth; principles of supersaturation; types of industrial crystallizers; safety issues associated with drying of volatile hydrocarbons.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basic principles of mass transfer operations such as extraction, distillation, drying, humidification and dehumidification.
CO2 - Explain the principles of mass transfer operations such as extraction, distillation, drying, humidification and dehumidification.
CO3 - Apply principles related to extraction, distillation, drying, humidification and dehumidification.
CO4 - Analyse the problems related to extraction, distillation, drying, humidification and dehumidification.
CO5 - Justify the applications of the theory learned in industrial practice regarding extraction, distillation, drying, humidification and dehumidification.
CO6 - Design mass transfer equipment based upon the conditions given in the problem statement.

TEXT / REFERENCE BOOKS

1. Treybal, R. E. (2012). *Mass transfer operations* (3rd ed.). McGraw Hill Education.
2. Dutta, B. K. (2006). *Principles of mass transfer and separation processes*. Prentice Hall India Learning.
3. Narayanan, K. V. (2005). *Mass transfer: Theory and applications*. CBS Publishers.
4. Wankat, P. C. (1988). *Separations in chemical engineering: Equilibrium staged separations*. Prentice Hall.
5. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill Education.

24CH306P					Mass Transfer - 2 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the principles of liquid-liquid and solid-liquid operations.
- To learn the application of simple distillation, steam distillation and fractional distillation.
- To explain the application of distillation, extraction, drying and crystallization operations.
- To formulate and solve engineering problems involving drying, distillation, extraction, humidification and dehumidification.

LIST OF EXPERIMENTS

1. To study the effect of phase ratio on the extraction of acetic acid (dispersed phase) by ethyl acetate (continuous phase).
2. To find out the number of stages required for the extraction of acetic acid (dispersed phase) by ethyl acetate (continuous phase).
3. To study solid-liquid extraction in a batch process and find the effect of solvent temperature on the percentage recovery of oil from oil seeds.
4. To determine the yield of benzoic acid crystals.
5. To find out the critical moisture content of a given material and determine its equation for the constant and falling rate periods.
6. Study of a rotary dryer.
7. Study of a forced draft cooling tower.
8. To perform a differential distillation experiment and verify the Rayleigh equation.
9. To study vapour-liquid equilibrium using simple distillation.
10. To study the steam distillation process using turpentine oil as feedstock.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define distillation, extraction, crystallization, drying, humidification and dehumidification.
CO2 - Explain the extent of drying and cooling.
CO3 - Apply principles related to distillation, extraction, crystallization, drying, humidification and dehumidification.
CO4 - Distinguish the effectiveness of distillation and extraction.
CO5 - Determine the characteristics of a cooling tower and the equilibrium moisture content.
CO6 - Estimate the design parameters for distillation, crystallization, extraction, drying, humidification and dehumidification.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Mass transfer laboratory manual*. Pandit Deendayal Energy University.
2. Treybal, R. E. (2012). *Mass transfer operations* (3rd ed.). McGraw Hill Education.
3. McCabe, W. L., Smith, J. C., & Harriott, P. (2005). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill.
4. Dutta, B. K. (2006). *Principles of mass transfer and separation processes*. Prentice Hall India Learning.
5. Narayanan, K. V. (2005). *Mass transfer: Theory and applications*. CBS Publishers.
6. Wankat, P. C. (1988). *Separations in chemical engineering: Equilibrium staged separations*. Prentice Hall.

24CH334T					Piping Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 2)

COURSE OBJECTIVES

- To create, select and apply appropriate modern software tools for fluid flow and pipeline engineering.
- To identify and learn the usage of proper fluid moving machines and pressure changers in a process plant.
- To learn to solve pipe flowsheeting problems more quickly, efficiently and successfully using computer-aided tools.
- To understand pressure drops to maximize flow and mitigate the risk of flow assurance issues, and to analyse piping networks along with process equipment.

Unit I: PIPE FLOW HYDRAULIC ANALYSIS AND DESIGN **Hours: 12**

Introduction to piping engineering; basics of fluid flow through pipes (Newtonian and non-Newtonian fluids); codes and standards for piping engineering and design; piping elements viz. pipes, fittings, flanges, gaskets, bolting and valves; types of valves; piping drawing layout and instrument diagrams.

Unit II: PRESSURE CHANGERS **Hours: 9**

Handling fluid streams to change their pressure for different reasons; pressure changers; pumps, compressors and valves - design and analysis using computational software; selection and efficiency.

Unit III: PIPING NETWORK ANALYSIS **Hours: 9**

Head losses due to contraction and expansion; other types of losses; network analysis; overall loss estimation through network analysis; optimizing piping networks with respect to losses; Hardy-Cross method; dynamic multi-phase pipeline modelling; integration of pipeline models with processing facilities to conduct what-if analyses and optimizations.

Unit IV: INDUSTRIAL PIPING SYSTEM **Hours: 9**

Piping for oil and gas industries; underground piping; design of water handling piping systems; sizing for equal velocity; sizing for equal areas; optimal sizing; water hammer; steam piping design; steam traps; slurry and sludge piping.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand, create, select and describe the design, codes, standards and specifications in piping engineering.
CO2 - Understand the design and characteristics of different industrial valves.
CO3 - Estimate overall pressure drop and maximize flow by the application of pipe fittings and pressure changers in piping system networks.
CO4 - Determine flowsheeting solutions by using design specification and sensitivity analysis approaches.
CO5 - Solve process design and simulation calculations of various unit operations using computational software.
CO6 - Outline process intensification and construct piping system networks.

TEXT / REFERENCE BOOKS

1. Haydary, J. (2019). *Chemical process design and simulation*. John Wiley & Sons.
2. Bausbacher, E., & Hunt, R. (1993). *Process plant layout and piping design*. PTR Prentice Hall.
3. Sahu, G. K. (1998). *Handbook of piping design*. New Age International Publishers.
4. Menon, E. S. (2005). *Piping calculation manual*. McGraw-Hill Education.

24CH335T					Corrosion Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 2)

COURSE OBJECTIVES

- To introduce the principles of the chemistry behind corrosion and the essential elements of electrochemical corrosion.
- To lay a foundation for understanding the forms of corrosion, corrosion mechanisms and electrochemical methods.
- To analyse the vulnerability of engineering systems to corrosion and the trailing losses.
- To develop the engineering methods to prevent and control various forms of corrosion.

Unit I: INTRODUCTION AND FUNDAMENTALS OF CORROSION **Hours: 12**

Definitions of corrosion; overall classification of types of corrosion; basic electrochemistry; galvanic and electrolytic cells; potential measurements; galvanic corrosion; bimetallic contacts; Eh-pH diagrams; electrode kinetics; polarization phenomena; polarization techniques to measure corrosion rates; cost of corrosion; metallurgical properties influencing corrosion; general testing procedures.

Unit II: TYPES OF CORROSION **Hours: 9**

Uniform attack; galvanic corrosion; crevice corrosion; pitting corrosion; intergranular corrosion; selective leaching; erosion corrosion; stress corrosion; mechanisms; testing procedures.

Unit III: ELECTRO-CHEMICAL METHODS OF CORROSION PREVENTION AND CONTROL **Hours: 9**

Mixed potential theory; cathodic protection; stray current corrosion; anodic control; electrochemical coatings; speciality coatings and surface protection; accidents in real life due to corrosion phenomena; engineering properties desired from materials; study of corrosion-resistant alloys of iron, copper, nickel, zinc and aluminium.

Unit IV: MECHANO-CHEMICAL METHODS OF CORROSION PREVENTION AND CONTROL **Hours: 9**

Design; mechanical and non-electrochemical coatings; corrosion inhibition; passivity phenomena and development of corrosion-resistant alloys.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basics of corrosion and its various forms.
CO2 - Demonstrate the testing procedures for corroding systems.
CO3 - Apply the electrochemical and metallurgical aspects of combating the eight forms of corrosion.
CO4 - Inspect the corrosion and provide the correct prevention and control method.
CO5 - Evaluate the vulnerability of any material system to corrosion.
CO6 - Design a suitable system to withstand corrosion under given conditions.

TEXT / REFERENCE BOOKS

1. Fontana, M. G. (2005). *Corrosion engineering* (3rd ed.). McGraw-Hill Book Company.
2. Jones, D. A. (1996). *Principles and prevention of corrosion* (2nd ed.). Prentice-Hall.
3. Uhlig, H. H., & Revie, R. W. (2008). *Corrosion and corrosion control* (4th ed.). Wiley.
4. Roberge, P. R. (2008). *Corrosion engineering: Principles and practice*. McGraw-Hill.

24CH336T					Material Science & Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 2)

COURSE OBJECTIVES

- To impart the basic concepts of material science.
- To understand the various properties, corrosion and heat treatment of engineering materials.
- To understand the engineering requirements and selection of materials for various applications.
- To gain knowledge of how machine learning is implemented for material synthesis.

Unit I: CRYSTAL STRUCTURE AND CHARACTERIZATION OF ENGINEERING MATERIALS **Hours: 9**

Different categories of engineering materials; selection of materials; crystal structure and crystal geometry; structure of solids; methods of determining structures; imperfections in crystals; types of imperfection; point imperfection; characterization of crystal structure through XRD, SAED pattern of TEM, UV-Vis, FTIR spectroscopy and Raman spectroscopy.

Unit II: PHYSICAL PROPERTIES OF ENGINEERING MATERIALS **Hours: 12**

Mechanical, electrical and magnetic properties of materials; physical significance of dielectric permittivity, electrical conductivity, magnetic permeability, magnetic susceptibility and Curie point; deformation of materials; heat treatment techniques; corrosion; theories of corrosion; control and prevention of corrosion; economic impact of corrosion in process industries; consideration of corrosion in engineering design.

Unit III: FERROUS AND NON-FERROUS MATERIALS FOR PROCESS INDUSTRIES **Hours: 9**

Engineering materials: ferrous metals, iron and their alloys; iron and steel; iron-carbon equilibrium diagram; non-ferrous metals and alloys - aluminium, copper, zinc, lead, nickel and their alloys with reference to applications in chemical industries.

Unit IV: CERAMIC MATERIALS, NANOMATERIALS AND THEIR APPLICATIONS **Hours: 9**

Inorganic materials: ceramics, glass and refractories; organic materials: wood, plastics and rubber; advanced materials (biomaterials, nanomaterials and composites) with special reference to applications in chemical industries; nanoparticle synthesis; application of nanomaterials in biomedical engineering, wastewater treatment, catalysis-photocatalysis and environmental remediation; AI in advanced nanomaterial synthesis.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics such as internal structure, crystal geometry and imperfections of engineering materials.
CO2 - Explain various properties of materials such as mechanical, electrical and magnetic properties.
CO3 - Explain corrosion, remedial measures for corrosion and its economic impact.
CO4 - Examine various materials for their potential application in the chemical industry.
CO5 - Predict the choice of material in chemical engineering in the area of equipment design.
CO6 - Discuss several nanomaterials, their usage in diverse fields and AI-based material synthesis.

TEXT / REFERENCE BOOKS

1. Van Vlack, L. H. (1989). *Elements of materials science and engineering* (6th ed.). Addison-Wesley.
2. Hajra Choudhury, S. K. (2006). *Material science and processes*. Indian Book Distribution Co.
3. Callister, W. D. (2007). *Materials science and engineering: An introduction* (7th ed.). John Wiley & Sons.
4. Raghavan, V. (2004). *Materials science and engineering: A first course* (5th ed.). Prentice Hall of India.

24CH337T					Nano Technology & Energy Storage					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 3)

COURSE OBJECTIVES

- To understand the fundamentals of nanotechnology.
- To explain the nanoscale paradigm in terms of properties at nanoscale dimensions.
- To identify current nanotechnology solutions in design, engineering and manufacturing.
- To understand the use of nanotechnology in energy storage and other applications.

Unit I: INTRODUCTION TO NANOTECHNOLOGY **Hours: 12**

Introduction to nanotechnology; history of nanotechnology; nanotechnology perspective - world and India; nano size and properties; classification of nanostructures; defects in nanomaterials; applications of nanomaterials; properties of nanomaterials from bulk to nano; nanomaterial case studies demonstrating non-classical behaviour at the nanoscale in successful and emergent nanotechnologies.

Unit II: SYNTHESIS OF NANOMATERIALS **Hours: 9**

Synthesis of nanomaterials; bottom-up approach: self-assembly and self-organization, vapour phase deposition, colloidal, sol-gel and simple pyrolysis; top-down approach: miniaturization of smaller structures from larger ones such as milling, lithography and machining; synthesis by templating methods - soft templating and hard templating; hydrothermal and solvothermal processes; microwave-assisted synthesis.

Unit III: CHARACTERIZATION OF NANOMATERIALS **Hours: 7**

Detailed characterization techniques based on radiation-matter interactions and their analytical applications: transmission electron microscope, scanning electron microscope, atomic force microscope, scanning tunnelling microscope, X-ray diffraction, small angle X-ray scattering, dynamic light scattering, spectroscopies, infra-red spectroscopy and FTIR, UV-Vis spectroscopy, Raman spectroscopy, surface area analysis; safety during handling of characterization equipment.

Unit IV: NANOMATERIALS FOR ENERGY AND VARIED APPLICATIONS **Hours: 11**

Applications of nanomaterials to catalysis, food and agriculture, semiconductors and electronics, structural materials, water treatment and environment, medical devices, drug delivery, paints, textiles and defence equipment; nanotechnology for sustainable energy; materials for light emitting diodes, batteries, advanced turbines, capacitors and fuel cells; nanotechnology-enabled renewable energy technologies; energy transport, conversion and storage; nano, micro and meso scale devices.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of nanotechnology, its terminologies and unique properties.
CO2 - Explain in detail the physical and chemical methods for the synthesis of nanoparticles.
CO3 - Develop various nanomaterials by understanding the requirements of application and properties.
CO4 - Compare various techniques for nanomaterial characterization and interpretation of results.
CO5 - Evaluate the characterization results for standard nanoparticles.
CO6 - Discuss, build and propose novel solutions on the use of nanotechnology in varied applications.

TEXT / REFERENCE BOOKS

1. Kelsall, R., Hamley, I. W., & Geoghegan, M. (2005). *Nanoscale science and technology*. John Wiley & Sons.
2. Lindsay, S. M. (2009). *Introduction to nanoscience*. Oxford University Press.
3. Kulkarni, S. K. (2015). *Nanotechnology: Principles and practices* (3rd ed.). Capital Publishing Company.
4. Murty, B. S., Shankar, P., Raj, B., Rath, B. B., & Murday, J. (2013). *Textbook of nanoscience and nanotechnology*. Universities Press.

24CH338T					Membrane Processes					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 3)

COURSE OBJECTIVES

- To acquire in-depth knowledge of different types of membranes, materials and transport mechanisms.
- To understand various membrane fabrication techniques and their characterization methods.
- To learn membrane modules for effective application in different membrane-based processes.
- To develop the understanding of concentration polarization and membrane fouling.

Unit I: MEMBRANE OVERVIEW AND TRANSPORT THEORY **Hours: 9**

Historical development of membranes; membrane materials - polymeric, inorganic and liquid; isotropic membranes; anisotropic membranes; inorganic membranes; liquid membranes; pore flow model; solution-diffusion model; structure-permeability relationships.

Unit II: MEMBRANE FABRICATION TECHNIQUES AND MODULES **Hours: 9**

Membrane preparation: phase inversion, immersion precipitation, solution evaporation, track-etch method, sol-gel process, interfacial polymerization, dip-coating process, film stretching and template leaching; membrane modules - hollow fibre, spiral wound and flat sheet.

Unit III: MEMBRANE CHARACTERIZATION **Hours: 9**

Membrane characterization: morphology, top surface and cross-section structure, surface properties, hydrophilicity and hydrophobicity, porosity, water flux, permeability, pore size, FTIR analysis, solute rejection properties and protein adsorption property.

Unit IV: MEMBRANE PROCESSES AND FOULING **Hours: 12**

Reverse osmosis; nanofiltration; ultrafiltration; microfiltration; gas separation; membrane distillation; membrane bioreactors; concentration polarization in liquid separation processes; gel layer model; osmotic pressure model; boundary layer resistance model; membrane fouling; fouling control.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basic knowledge of membranes, materials, types and transport mechanisms.
CO2 - Illustrate skills in the areas of fabrication of isotropic and anisotropic membranes.
CO3 - Identify the different membrane modules and demonstrate skills in characterization techniques.
CO4 - Examine the mechanism of concentration polarization and membrane fouling.
CO5 - Assess transport models in membrane separation processes.
CO6 - Estimate the membrane performance and separation characteristics of different processes.

TEXT / REFERENCE BOOKS

1. Baker, R. W. (2012). *Membrane technology and applications* (3rd ed.). John Wiley & Sons.
2. Ismail, A. F., & Matsuura, T. (2019). *Membrane separation processes*. Elsevier.
3. Ismail, A. F., & Rahman, M. A. (2019). *Membrane separation principles and applications*. Elsevier.
4. Nath, K. (2017). *Membrane separation processes* (2nd ed.). PHI Learning.

24CH339T					Environmental Engineering and Pollution Control					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 3)

COURSE OBJECTIVES

- To understand environmental regulations and standards.
- To understand the principles and design of air pollution control devices.
- To discuss the characteristics and treatment methods of water and wastewater.
- To understand solid waste management and noise pollution control.

Unit I: ENVIRONMENTAL STANDARDS AND AIR POLLUTION CONTROL **Hours: 11**

Classification of pollutants and their permissible limits; air pollutant concentration calculations; air quality index; air pollution control: settling chambers, cyclone separators, dust collectors, fabric filters, venturi scrubbers, electrostatic precipitators, wet scrubbers, adsorption, absorption and catalytic reduction (e.g. SCR).

Unit II: PRELIMINARY AND PRIMARY TREATMENT PROCESSES **Hours: 10**

Sources of water; impurities in water; Indian and WHO standards for drinking water; water-borne diseases and their control; sources of wastewater; physical, chemical and biological characteristics of water and wastewater; water quality index calculation; preliminary treatment processes - screens and skimming; primary treatment processes - sedimentation, coagulation and flocculation, sand filtration.

Unit III: SECONDARY AND TERTIARY TREATMENT PROCESSES **Hours: 9**

Secondary treatment processes: activated sludge process - design procedures for HRT, F/M ratio, SVI, MLSS and sludge age; trickling filters and their efficiency - standard, high rate and two-stage; sludge treatment - sludge digestion process. Tertiary treatment processes: disinfection, membrane processes, adsorption and ion exchange, ozonation, aeration, softening, fluoridation, decarbonation, lime-soda softening and demineralization.

Unit IV: SOLID WASTE MANAGEMENT AND NOISE POLLUTION CONTROL **Hours: 9**

Solid waste management: quantity, composition and characteristics of solid waste; methods of solid waste treatment and disposal; treatment of plastic and e-waste. Noise pollution: levels and standards, effects and control.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define environmental regulatory legislation and standards and various types of pollution.
CO2 - Demonstrate different types of waste generated, disposal methods, reduction and monitoring processes.
CO3 - Solve design calculations for air pollution control devices.
CO4 - Analyse the properties of water and wastewater.
CO5 - Assess pollution control in air and treatment processes of water, wastewater and solids.
CO6 - Acquire knowledge about the various environmental and safety standards and legislation.

TEXT / REFERENCE BOOKS

1. Rao, C. S. (2007). *Environmental pollution control engineering* (2nd ed.). New Age International.
2. Peavy, H. S., Rowe, D. R., & Tchobanoglous, G. (1985). *Environmental engineering*. McGraw-Hill.
3. Metcalf & Eddy, Inc. (2014). *Wastewater engineering: Treatment and reuse* (5th ed.). McGraw Hill.
4. Mahajan, S. P. (1985). *Pollution control in process industries*. Tata McGraw Hill.